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Catalyst Cloud and New Zealand's Government Cloud Strategy

1 Abstract

In our report we have assessed the implementation of cloud computing in the New Zealand Government with the emphasis on the use of Catalyst Cloud as a part of the All-of-Government (AoG) Cloud Framework. Our analysis aims to determine the effectiveness with which the government uses cloud services to improve the digital transformation, operational resilience and service delivery.

The NZ Government as the main user will be working in a controlled system to guarantee compliance, data sovereignty, and interoperability between agencies and departments through a designated Department of Internal Affairs (DIA).

Catalyst Cloud, which is an infrastructure-as-a-service (IaaS) provider, is operated in New Zealand and has three data centres within the country, which meet the national data-residency and privacy requirements.

The results point to significant advantages including increased scalability, increased data sovereignty, and increased business continuity. Nevertheless, there are still difficulties associated with integrating the cloud, upskilling the workforce, and cost optimization across the legacy systems.

Altogether, the adherence of Catalyst Cloud to the open-source and government-sanctioned models proves its ability to integrate secure, sovereign, and resilient cloud adoption within the public sector, which makes NZ more dedicated to digital trust, transparency, and local innovation.

2 Introduction

Cloud computing has emerged as an essential part of the digital transformation in the private and the public sectors. It is a demand-based provision of computing services like servers, storage, databases, networking and analytics in the form of the Cloud by a pay-as-you-go model [6]. According to the U.S. national institute of standards and technology (NIST), there are three models of cloud computing [43]:

- **Infrastructure as a Service (IaaS)** – provides virtual computing capabilities such as storage and servers, allowing users to manage and deploy their own systems and applications.
- **Platform as a Service (PaaS)** – offers a development and deployment platform where applications can be built, tested, and hosted without requiring developers to manage the underlying infrastructure.
- **Software as a Service (SaaS)** – delivers fully managed software applications over the Internet, enabling end-users to access services without maintaining on-site hardware or software [44].

The combination of these service models facilitates flexibility, scalability, cost-effectiveness, and as a result, governments can modernise legacy systems and enhance data resilience, provide citizen-focused services effectively. In the case of the public organisations, cloud computing also provides business continuity and disaster recovery because it offers data replication and redundancy across environments.

The government in New Zealand identifies cloud computing as an agent of digital innovation and agility in the delivery of services to the population. The Cloud First Policy that has been proposed by the Department of Internal Affairs (DIA) points to the fact that the agencies are supposed to assume that public cloud solutions are the default choice when considering new investments in ICT where security and privacy considerations can be achieved [35]. This policy will be in line with the Digital Public Service Strategy 2025, which aims at having a trusted and responsive digital government that will provide secure, integrated, and people-centric digital services [37]. To realise this vision, an All-of-Government (AoG) Cloud Framework was created to facilitate agencies in procuring, deploying and managing cloud services in a secure manner [34].

NEW ZEALAND GOVERNMENT:
CLOUD FIRST FOR A DIGITAL PUBLIC SERVICE

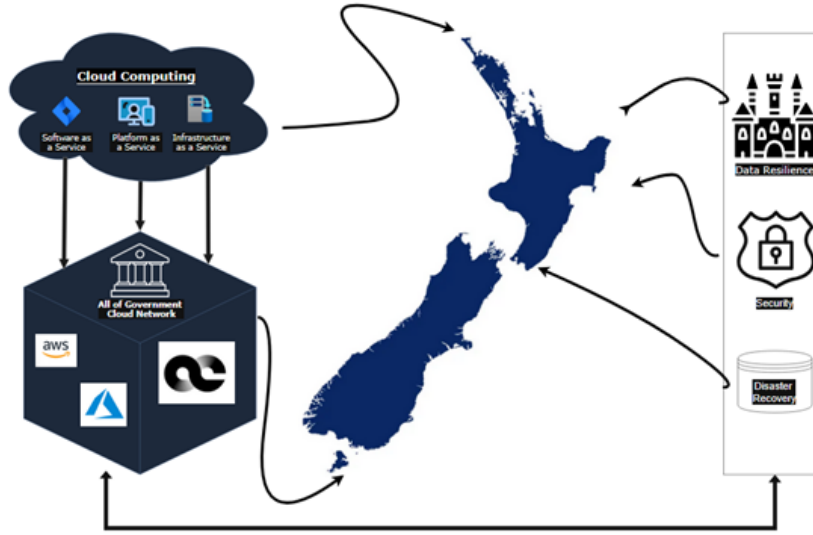


Figure 1: New Zealand: Cloud first for a Digital Public Service

In this regard, our report will assess the implementation of cloud computing by the New Zealand Government, which requires the use of Catalyst Cloud services to manage its operational and continuity based on business continuity, disaster recovery, security, and governance. The purpose of the analysis is to find out the contribution of Catalyst Cloud to the AoG Cloud Framework.

3 Organisation Background

To provide the required services in the fields of policy, healthcare, education, transport, and infrastructure, the NZ Government has a complex system of the public sector organizations. These statutorily controlled organizations that are accountable to ministers pursuant to the Public Service Act 2020 [27] consist of ministries, departments, crown entities, and district health agencies. They combine to create a dependent network that is becoming more dependent on digital technologies to provide the advantages of the state, enhance the life of the residents and sustain its functioning. The Ministry of Health (Manatu Hauora) is one of the NZ organizations to which most data is used in this respect. It manages the national health policy regulation and coordinates the provision of treatment by the hospital and regional networks. The Ministry needs to have a secure, compliant, and robust digital infrastructure because it deals with a great number of sensitive health information, including patient records, epidemiological data, and emergency response systems. The growth of data-driven

healthcare has brought about the need to use powerful cloud technology to store, analyze, and preserve business continuity within the Ministry [33].

Department of Internal Affairs (DIA) is one of the key participants when it comes to the digital transformation plan by the government while issuing frameworks on procurement, assurance and policy on the ICT services in the public sector [36].

The DIA sets agency standards of cybersecurity, privacy compliance, and interoperability of digital services through the Government Chief Digital Officer (GCDO). To institutionalize the process of migrating to the cloud and create a trusted supplier landscape, the DIA developed the All-of-Government (AoG) Cloud Framework upon recognizing the difficulties of having a haphazard procurement and unequal security processes [35].

Catalyst Cloud Ltd., The company is the first and the only domestic-owned public cloud company in NZ that operates three data centers based in Wellington, Hamilton, and Porirua [8].

The entire infrastructure of the company is developed on open-source technologies, including Ceph (storage), Kubernetes (container), and OpenStack (virtualization) [21]. This architecture is not vendor-locked, which encourages transparency, interoperability and flexibility in accordance with the New Zealand government requirements of the importance of open standards and independence. AoG Cloud Framework formally listed Catalyst Cloud as a service provider, implying that the government agencies may purchase its services based on the pre-negotiated contract which meets the ISO 27001 security certification and NZISM requirements [24].

Catalyst being a sovereign cloud provider ensures that all the information, storage and the backup are only done within the borders of NZ. Catalyst Cloud is the service provider, which has IaaS and PaaS service providers including data hosting, analytics, and disaster recovery services.

The NZ Government has been discovered as the customer of this arrangement and buys services through the AoG service framework of the DIA. As per the requirements of resilience and data security to the public sector, the ministry of health uses the infrastructure proposed by Catalyst to make sure that crucial health services are operational during a national disaster or cyber attack [30].



Figure 2: Structure of cloud governance under the New Zealand Government

4 Current Cloud Setup and Architecture

Catalyst Cloud owns data centers in New Zealand and offers common cloud services, such as IaaS, SaaS, and PaaS, built on an open-source platform called OpenStack.

4.1 Cloud Service type

The Catalyst Cloud Framework Agreement, signed between the NZ government and Catalyst Cloud, includes IaaS and PaaS [32]. There is a healthcare web service called My Health Record, provided by Health New Zealand. On this website, users can view and manage their personal health history, including immunisation and COVID-19 test results [41]. For a cloud service to support a website, PaaS for web hosting, such as AWS Elastic Beanstalk and Azure web apps, can be one of the powerful solutions for web deployment. However, as of now, Catalyst Cloud does not provide this type of PaaS. So, it could be a virtual machine on the cloud, as an IaaS model is used for hosting the website on it. In fact, web hosting is one of the major uses of the virtual machine on the cloud [29].

4.2 Data Centre location



Figure 3: Data Centers for Catalyst Cloud Inside New Zealand

Catalyst Cloud has three data centres in New Zealand for data resiliency, with a WAN network connection between them, as other cloud providers do. Each of them is in Porirua, Wellington, and Hamilton [17]. According to the paper [7], the physical distance between the data centre and the consumer directly impacts the QOS of the

network. The main office for the Ministry of Health and other government-related offices is primarily located in Wellington, making it a suitable location for government organisations. However, the distance between Porirua and Wellington is only about 22 km. Ideally, each data centre should be built further apart, because it cannot serve as a measurement of data resilience. Especially for earthquakes that frequently happen in NZ.

4.3 Open-Source Based architecture

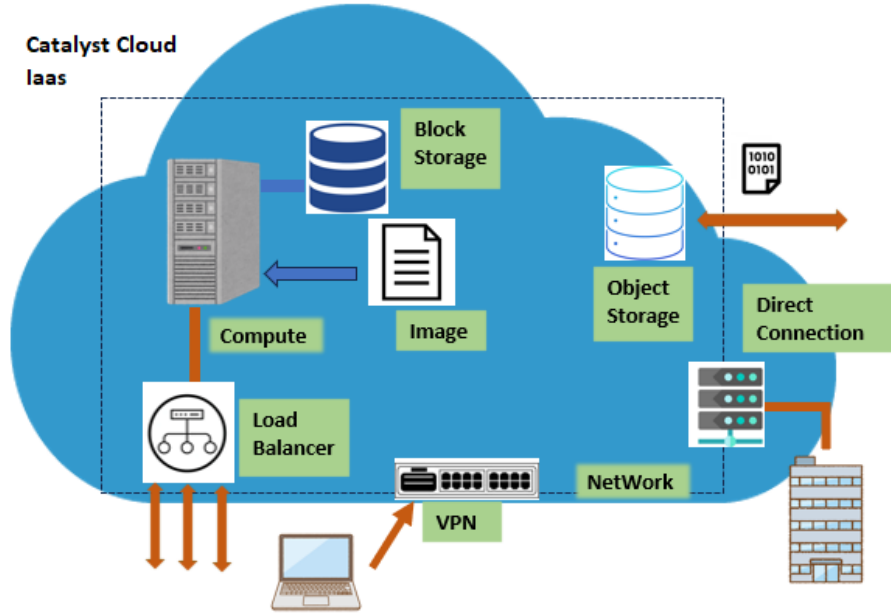


Figure 4: IaaS provided by Catalyst Cloud

Catalyst Cloud is built on the open-source OpenStack [15], which is used by various organisations, including the United Nations and Walmart [40]. Interestingly, this paper [29] indicates that OpenStack has slightly better performance than the Google Cloud Platform based on the various benchmarks. It implies that OpenStack is as good as other major cloud services. Additionally, the paper [47] demonstrates that OpenStack exhibits the highest resiliency among the three major IaaS private cloud open-source platforms, which is particularly valuable for government organisations, as data resiliency is the most important task for them to keep the system and data safe, rather than providing fancy commercialised services. As mentioned previously, Catalyst Cloud does not offer a web hosting function. It could be because OpenStack does not have such a function. It means that the functions in the Catalyst cloud highly depend on OpenStack. One example is that the even dashboard screen in Catalyst Cloud

appears to be the same as the OpenStack dashboard, with some minor customisations, as shown in the following images [39] [12].

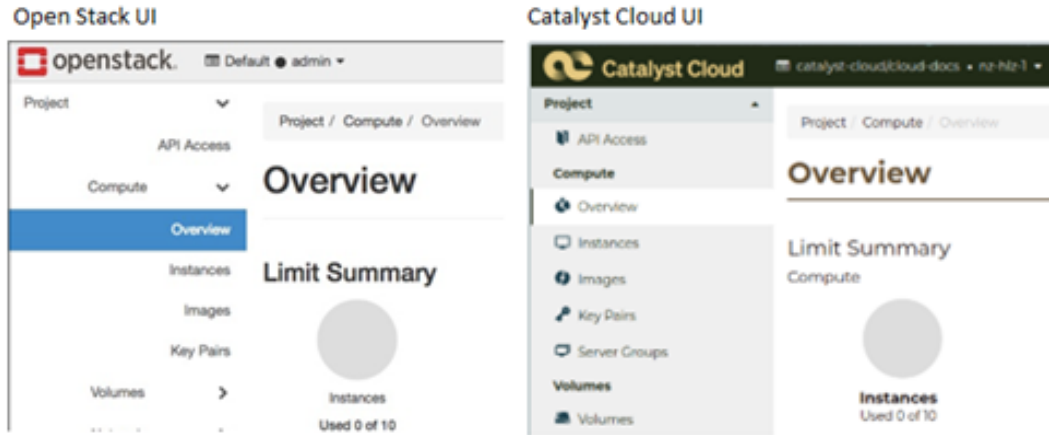


Figure 5: UI comparison between Open Stack and Catalyst Cloud [39] [12]

5 Benefits of Adopting Catalyst Cloud

There are several benefits for New Zealand public agencies to adopt Catalyst Cloud. Also, they make significant contributions to New Zealand’s economy and environment.

5.1 Data Sovereignty

One of the primary benefits of adopting Catalyst Cloud for government agencies in New Zealand, including the Ministry of Health, is data sovereignty. Data sovereignty encompasses not only storing data locally but also managing data in accordance with local laws. As the article [9] mentions, cloud services run by US companies, such as Amazon AWS and Microsoft Azure, are subject to the US Cloud Act, which requires them to provide data to the US government if ordered to do so, even if their data centre is located in New Zealand. Confidential data, including personal information or any security-related data, must not be compromised under any circumstances. Catalyst Cloud is a New Zealand company that runs the cloud service with data centres located in New Zealand, employing members under New Zealand law, and data is governed by New Zealand law [18]. Therefore, it is the ideal condition in terms of data sovereignty for New Zealand, which is why the New Zealand government prefers this local cloud provider.

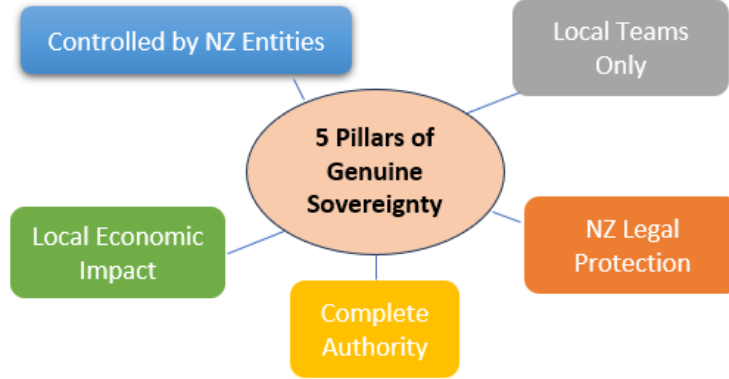


Figure 6: 5 pillars of Genuine Sovereignty under cloud catalyst

5.2 Local Economy Contribution

Recently, RNZ reported that the data centre generates an annual profit of \$16.5 billion for New Zealand, [47] equivalent to 6% of the country's GDP in 2024. Catalyst Cloud is a key component in its profit generation. The data centre contributes to the country's economy in various fields, including employment growth and the overall digital logistics chain. Although there will be some challenges, such as power generation and finding skilled human resources [48].

5.3 Environmental sustainability

The report [28] revealed that U.S. data centres in 2013 consumed a vast amount of electricity, which is equivalent to the annual output of 34 large (500-megawatt) coal-fired power plants. This is enough power for all the households in New York City twice over. Nowadays, AI technologies are commonly used, and they consume huge PC resources. Therefore, the power requirement is expected to be significantly higher than in 2013, which would lead to a negative environmental impact due to the increased carbon footprint. Additionally, electricity consumption is anticipated to more than double from 2024 to 2030, as shown in the graph [5]. Catalyst Cloud participates in the Net Zero Carbon initiative and has renewed its carbon reduction certification, called Toitū, this year. It encourages other public sector organisations required to meet carbon neutrality targets by 2025 [10]. Several measures can be taken to reduce carbon emissions in data centres, including the use of green energy sources such as solar power and hydroelectric power, as well as innovations in cooling technology and computer architecture [42]. Catalyst Cloud utilises 100% renewable electricity sources to power its data centre [10].

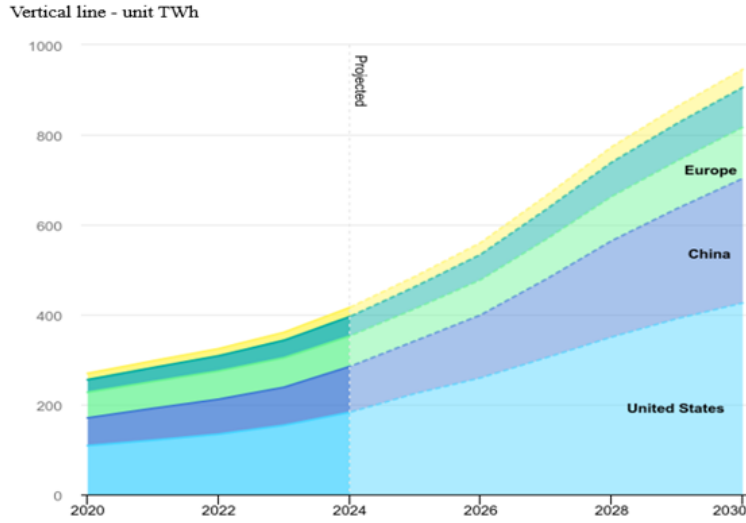


Figure 7: Data centre electricity consumption by region [5]

6 Challenges and Limitations

Due to local ownership of services, several challenges arise, including the employment of high-skilled engineers for cloud services and the lack of global redundancy.

6.1 Integration with Legacy Systems

Many health workloads are still tied to tightly coupled applications and on-premises databases. When these systems are replaced or tightly integrated with cloud services, NZ government guidance calls for case-by-case assessment, jurisdictional risk analysis, and formal accreditation [3]. That slows migration and forces deeper thinking about interface design, data classification, and mapping controls to PSR/NZISM [4].

6.2 Skills gap training and education for cloud operations & DevOps

Many public agencies lack the in-house capability to securely run Kubernetes, infrastructure-as-code, and CI/CD in a regulated environment. Success with DevOps depends on competent tooling, genuine collaboration, and automated testing; shortfalls in skills and maturity remain common barriers[38]. Catalyst Cloud offers managed services [16], but MoH would still need to fund platform/SRE uplift, secure pipelines, and policy-as-code guardrails.

6.3 Hyperscale feature set, limited global scale

Catalyst Cloud prioritises NZ sovereignty and locality with three regions and a curated service catalogue [20]. That’s positive for jurisdictional control, but it’s not as broad as AWS or Azure in terms of global edge coverage and the depth of managed

services. Agencies may need to “build over buy” using open-source stacks on Kubernetes and accept fewer pre-built compliance artifacts than those available in hyperscale ecosystems.

6.4 Budget management & cost visibility

Without disciplined tagging, per-product budgets, storage lifecycle rules, and reserved-capacity planning, containerised and event-driven estates can quickly create “cost fog.” DIA recommends making risks, controls, and costs explicit and maintaining a living risk register pair that with FinOps guardrails (budgets, alerts, and showback/chargeback) to prevent cost creep and shadow IT [3].

6.5 Resistance to change and change management

PSR and NZISM both stress early security engagement and governance alignment [4]. Where roles and processes are unclear or controls feel inflexible, the security function can be perceived as friction by delivery teams; cultural resistance and anxiety about automation clash with research showing faster outcomes from DevOps adoption. Leadership-driven operating-model shifts (product teams, a platform team, explicit RACI) plus formal training are essential [38].

7 Business Continuity and Disaster Recovery (BCDR)

Like the major IT cloud companies, Catalyst Cloud performs measurements for business continuity and data protection. This is even more important in New Zealand prone to earthquakes.

7.1 Regional footprint and continuity

Catalyst Cloud’s sovereign, multi-region footprint supports continuity planning for health workloads. Its operational regions, Porirua, Hamilton and Wellington provide geographic isolation for fault tolerance. Access to the Auckland peering exchange also improves reachability and interconnect options for users [17]. This distributed design lowers single-site risk and aligns with continuity objectives in agency security risk management plans under the Protective Security Requirements (PSR) for state services that require resilience [31].

7.2 Data protection and platform resilience

Catalyst Object Storage is a masterless, distributed service that asynchronously replicates data to maintain three copies across different regions, and it runs regular integrity checks (e.g., CRC) to detect bit rot. This supports rapid recovery from disk, server, or regional loss [19]. Managed Database streamlines replica creation and promotion/replication workflows, improving availability for stateful services; point-in-time restoration is supported via automated and incremental backups [26]. More broadly, common backup patterns (e.g., scheduled jobs to Object Storage) can be

automated to meet retention and recovery SLAs [38].

7.3 Government Continuity Requirements

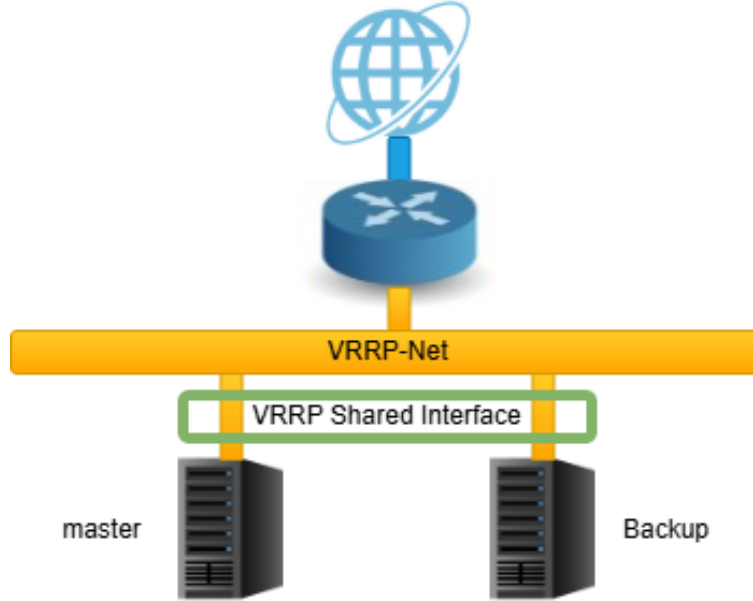


Figure 8: Redundancy example using VRRP in Catalyst Cloud

Under the PSR, agencies must embed resilience across physical, personnel, and information security. They need to set availability and recovery targets and deploy controls, including backup and recovery capabilities aligned to their security risk management plans and the NZISM [31] [4]. Together, these frameworks drive clear service tiers, realistic test cadences, and concrete proof points for recoverability. Catalyst Cloud is capable of VRRP (Virtual Router Redundancy Protocol), which provides high availability through keep-alive mechanisms. It can contribute to business continuity [13].

7.4 RTO/RPO in practice

RTO is the maximum acceptable downtime before business impact becomes unacceptable. RPO is the point in time to which data must be restored after an outage [46]. For example, public-facing APIs with high availability SLAs might target RTO 1 hour and RPO 15 minutes via active/standby multi-region deployments with continuous replication. Internal analytics portals might tolerate an RTO of 4–8 hours and an RPO of 1–4 hours using scheduled backups and replica promotion.

7.5 Local cloud vs on-prem BCDR

Multiple NZ regions, low-latency user access, and built-in HA/DR features in services like Kubernetes, databases, and storage help a local cloud accelerate time to a defensible continuity posture without the single-data-center risk [17] [19] [26] [3]. On-prem BCDR usually requires duplicate facilities, power, cooling, and network paths, with slower procurement and testing cycles; matching the same geographic diversity and operational maturity can be prohibitively expensive for a single agency.

7.6 Example scenarios

7.6.1 Major regional incident (earthquake impacting Wellington):

Scale up the standby deployment and shift web/API traffic to Hamilton or Porirua. Promote the database replica, validate object replication lag, then cut over DNS/load balancer [17] [26].

7.6.2 Cyberattack (ransomware) on an application tier:

Isolate affected workloads, restore the application tier from incremental backups or a warm standby cluster, rotate secrets, verify the last good RPO, and confirm RTO objectives were met [38] [46].

Planned DR exercises should verify the end-to-end promotion, cutover, and data-integrity checks, capturing evidence to meet PSR/NZISM assurance requirements [31] [4].

8 Security and Governance Considerations

Ensuring strong security and effective governance is crucial when adopting cloud computing in the New Zealand public sector.

8.1 Security policies

Cloud computing implementations must comply with national frameworks such as the New Zealand Information Security Manual (NZISM) and the Protective Security Requirements (PSRs), which together define the minimum requirements for protecting government information and systems [31] [4]. The Catalyst Cloud, which has certified regions in Porirua, Hamilton, and Wellington, have demonstrated compliance with PCI DSS, ISO 27001, and ISO 27017, ensuring that security best practices are consistently adhered to across their entire infrastructure [17] [23] [45].

Code	Name	PCI DSS	ISO 27001	ISO 27017
nz-por-1	Porirua	Certified	Certified	Certified
nz-hlz-1	Hamilton	Certified	Certified	Certified
nz_wlg_2	Wellington	Certified	Certified	Certified



Figure 9: Compliance and Security certifications of each region [17]

8.2 Data protection

Data protection is the cornerstone of reliable cloud operations. Information must be encrypted at rest and in transit using strong encryption standards to prevent unauthorized disclosure [45]. Access controls enforce the principle of least privilege, ensuring users only access the data required for their roles [4]. Catalyst Cloud’s published controls demonstrate how managed encryption keys, logging, and real-time monitoring protect customer and organization data across on-premises data centres [23].

8.3 Identity and Access Management (IAM)

Effective identity and access management (IAM) is crucial for securing multi-tenant environments. Implementing role-based access control (RBAC) allows administrators to assign permissions based on job function, reducing the risk of privilege abuse [2].

8.4 Audit and Compliance Reporting

Audit and compliance reporting mechanisms further enhance accountability. Regular internal and external audits ensure compliance with NZISM and ISO 27001 controls [45] [4], while automated logs provide traceability of user actions and configuration changes [23]. These reports are crucial for demonstrating due diligence during compliance reviews or incident investigations [2].

8.5 Governance Framework

A comprehensive governance framework is fundamental to the security of public sector cloud operations. The Department of Internal Affairs (DIA) provides central oversight and policy guidance, while agency Chief Information Officers (CIOs) are responsible for operational compliance and risk management within their organizations [2]. Additionally, the Digital.govt.nz Cloud Guidelines emphasize risk assessment, vendor evaluation, and assurance processes to maintain alignment with public sector security principles [2].

8.6 Transparency

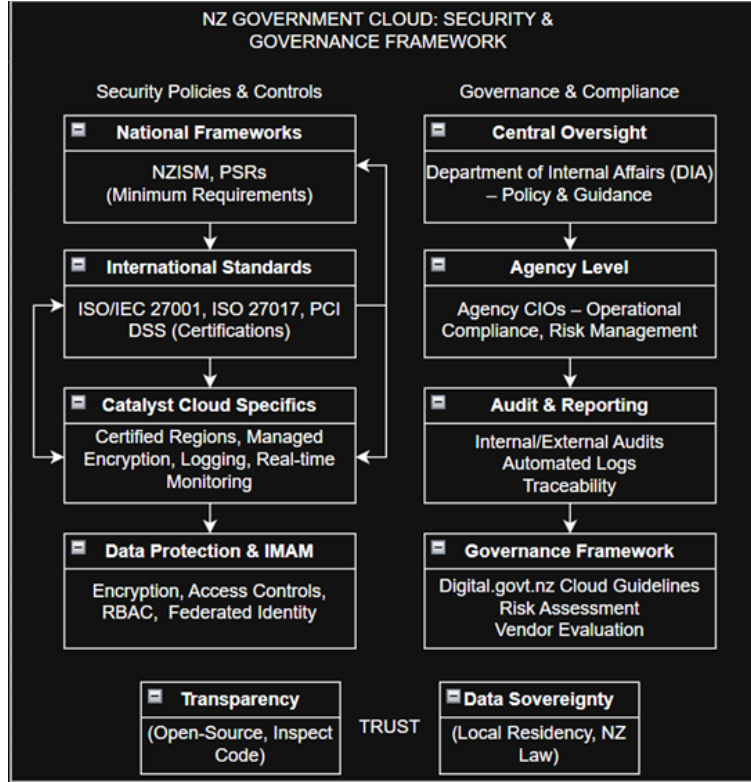


Figure 10: Security and Governance Framework

Transparency and trust are further enhanced through open-source technology and local data sovereignty. Catalyst Cloud’s open-source stack allows institutions to inspect source code, reducing the risk of hidden vulnerabilities and facilitating community validation. Local data residency within New Zealand ensures compliance with privacy laws and strengthens citizen confidence that sensitive data is governed by its jurisdiction [23] [45].

In summary, New Zealand’s approach to cloud computing ensures public sector cloud environments are secure, resilient and trusted by adhering to national frameworks, international standards and transparent governance practices.

9 Cost and Scalability

Adopting a cloud model shifts the financial burden from capital expenditures (CapEx) to operating expenditures (OpEx). Instead of investing in on-premises hardware and maintenance, businesses pay only for the computing resources they use. This pay-as-

you-go model transforms fixed infrastructure costs into variable expenses that align with actual usage, providing greater budget flexibility for public institutions with fluctuating workloads [22] [1].

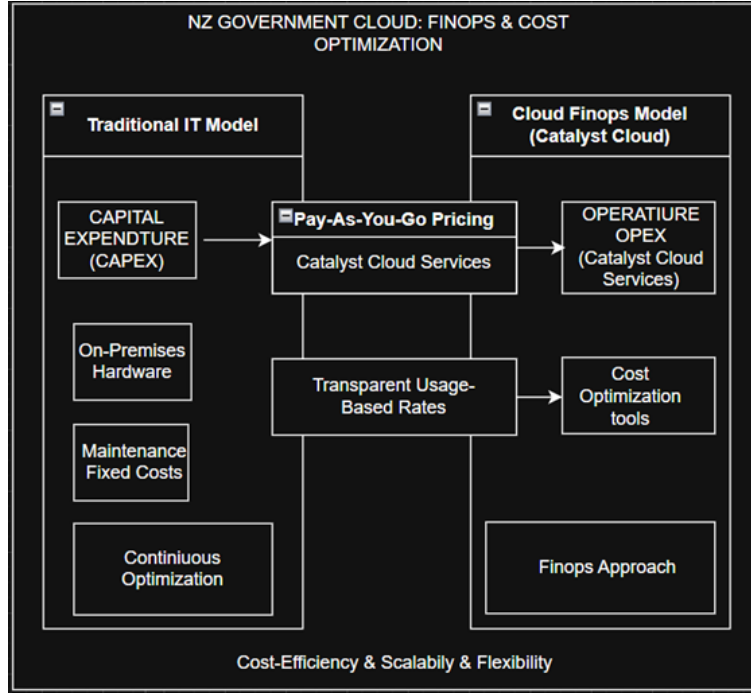


Figure 11: FinOps and Cost Optimization

Catalyst Cloud is a New Zealand-based provider with fully transparent, usage-based pricing. Each service (compute, storage, network) is priced per unit, and customers have access to public price lists showing hourly and monthly rates [22]. Catalyst Cloud also provides a cost calculator that estimates expected monthly expenses based on virtual CPU, memory, and storage requirements [11]. In addition, discount programs allow customers to optimize costs through regular commitments or high-volume usage [14]. Together, these tools support a FinOps (financial operations) approach, promoting accountability and continuous optimization of cloud spending.

From an operational scalability perspective, cloud computing provides dynamic elasticity. Agencies can automatically increase or decrease resources based on demand (e.g., during peak tax season or increased traffic on e-government portals) without having to purchase additional hardware [1]. This elasticity ensures performance reliability and cost efficiency, as agencies only pay for the capacity they use. However, this same flexibility also carries the risk of cost overruns when unused re-

sources remain active. Leaving instances, storage volumes, or test environments constantly running can quickly accumulate unnecessary expenses. To mitigate this, Catalyst Cloud provides usage cost monitoring dashboards that visualize spending per service and detect anomalies over time [25]. Regularly reviewing these reports, combined with automated shutdown policies, helps maintain financial discipline and align cloud spending with operational goals.

In summary, cloud computing adoption enables public agencies to achieve cost efficiency, scalability, and flexibility by leveraging consumption-based pricing and elastic capacity [1]. However, successful financial management depends on ongoing monitoring, governance, and optimization to ensure that cloud investments deliver measurable business value [22] [25].

10 Potential Issues and Recommendations

Even though the New Zealand Government has made good progress under the All-of-Government (AoG) Cloud Framework, some challenges still slow down the full use of cloud technology in the public sector.

The first challenge is the skills gap [38]. Many government agencies do not have enough people with the right cloud and DevOps skills to manage systems like OpenStack and Kubernetes. These open-source systems require advanced technical skills, which many government departments still lack. The Digital Public Service Strategy 2025 and the Cloud Adoption Policy and Strategy both stress the importance of building cloud skills to ensure safe and effective digital transformation [1] [4] [37]. To address this, the government could set up a Public Sector Cloud Academy to train employees in cloud operations, cybersecurity, and data protection based on PSR and NZISM guidelines [31] [4].

Another issue is integration complexity. Many ministries still depend on old systems that are hard to connect to modern cloud platforms [31] [35]. A hybrid migration approach can help by linking old on-premises systems with Catalyst Cloud’s IaaS and PaaS using secure VPNs [24] [21]. This allows a smooth and gradual move to the cloud without disrupting important services.

Cost tracking is also a concern. Because cloud resources can grow or shrink quickly, it can be hard to see where money is being spent. Without good tracking, agencies risk overspending [25] [22]. A central FinOps dashboard that uses Catalyst Cloud’s pricing and usage tools [25] [22] [14] would help all agencies monitor spending, set budgets, and improve transparency.

There are also governance issues. Different agencies sometimes apply cloud and security rules like PSR and NZISM in different ways, leading to confusion and duplication [31] [4]. Creating a Cross-Agency Cloud Governance Committee under the Department of Internal Affairs would help align policies, share good practices, and improve coordination [1] [35].

Lastly, vendor diversification is important. While Catalyst Cloud supports data sovereignty,

depending on only one provider can limit innovation. Using a multi-cloud approach with AWS or Azure [1] [22] [34] can balance local control with access to global technology. It would improve system reliability. It can get used as part of the AoG Framework.

By addressing these issues, the New Zealand Government can build stronger cloud skills, manage costs better, improve security, and deliver modern, reliable digital services for citizens.

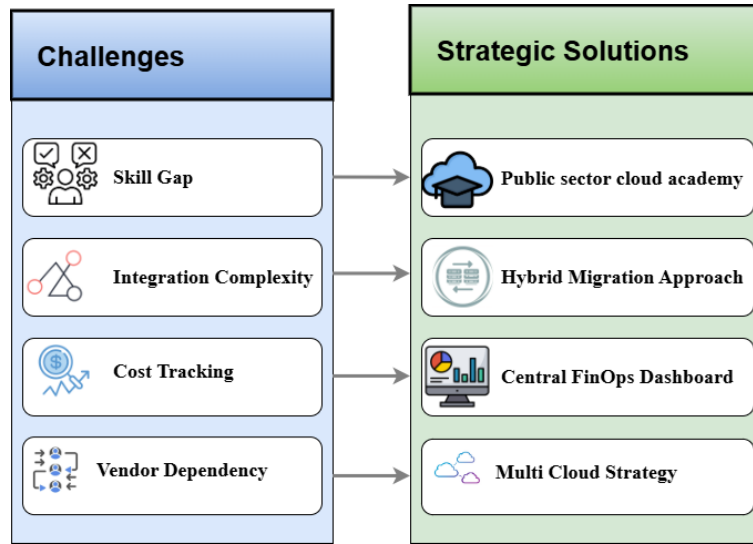


Figure 12: Challenges and solutions for full adoption

11 Conclusion

The All-of-Government (AoG) Cloud Framework has played an important role in helping New Zealand’s public sector move toward digital transformation. The government is getting benefits of local data storage, strong security standards, and improved business continuity by using Catalyst Cloud. Its open-source and renewable infrastructure which also supports transparency and sustainability of goals.

However, there are still some problems. Many staff need better cloud skills, older systems are hard to connect, cost tracking can be unclear, and different agencies often follow different rules. Creating a Public Sector Cloud Academy, using hybrid cloud migration, improving cost monitoring, and setting up a cross-agency team can help solve these issues.

By also using a multi-cloud approach with providers like AWS or Azure, the government can keep control of local data while gaining access to the latest global tools. These steps will help New Zealand build a secure, efficient, and future-ready public

cloud system.

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Reflection:

1. Ravi Ramchandani (1605492)

During the group assessment we had a goal to assess the adoption of cloud computing in organisations in New Zealand and we decided as a team to look into the New Zealand government and search for organisations that are involved in the development of this technology. My Main contribution was to research, write and formulate the introduction of cloud computing technologies used, organizational background of the companies involved and also the understanding of the cloud computing infrastructure, the nature of the digital governance environment in New Zealand, and lastly the role played by Catalyst Cloud in the operation of the All-of-Government (AoG) Cloud Framework.

We did by creating the analytical foundation of the report as the synthesis of the academic sources, government documents, policy frameworks, and technical specifications enabled us to establish the analytical foundation of the entire report. My significant input was making clear the role of the Department of Internal Affairs (DIA) and its influence on the formation of the national cloud strategy by the Cloud First Policy, cybersecurity standards, and procurement guidelines. I also looked at the value of data sovereignty that became a central theme in explaining the reason why the New Zealand Government is increasingly using a provider based locally like Catalyst Cloud.

Then we found the government organization Ministry of Health which has been rendering the services of Catalyst Cloud then based on this we were able to illustrate how the use of secure, compliant, and resilient cloud services can be beneficial to the work of the agencies of the public sector. Though we faced a number of challenges associated with the project. One, it was necessary to align various sources: academic literature, governmental policy documents, etc. and make sure that it is accurate and consistent. Our team divided work in different sections and made our documentation in an organized manner by equally distributing work amongst each other and when we started to combine our works in different sections there was a direction since we had multiple meetings to have our document and presentation organized. We had some challenges in creating the visuals and designs since not all are specialized in that area, as a team we worked together and made things work. Also communication between all sections was necessary to coordinate members contributions and prevent duplication and conceptual drift. Out of these issues, key lessons were learnt. In the end it was a very notable experience working together as a team.

In retrospective, this project enhanced my research and writing capabilities as well as abilities to analyze and my knowledge of how cloud computing aids transformation in New Zealand in the state sector.

2. Falak Share (1311616)

My contribution to this group project covered the “Challenges and Limitations” and the “Business Continuity and Disaster Recovery (BCDR)” sections. For the presentation PowerPoint, each of us completed our own part, and I then converted mine into a clear poster, using diagrams and small shaded bullet points to keep the ideas short and easy to follow. This also meant matching my content to our shared poster style.

The hardest part for me was turning technical ideas into plain English. At first, I wrote far too much detail about legacy systems, skills gaps, and multi-region failover. Because of that, both the slides and the poster were packed up to, and maybe past, their word and space limits. It was difficult to cut more text without losing the key messages. I also needed to think carefully about the layout, font size, and colours so the overall design looked tidy instead of cluttered.

Overall, working with the group was a very rewarding experience. Everyone was helpful, open to discussion, and willing to share feedback. I enjoyed seeing how our sections connected with each other. On a broader level, this project improved my understanding of cloud risk principles, BCDR, PSR/NZISM, and how to explain technical topics to non-technical people.

3. Masaru Yoshikawa (1595332)

It was great to have a group with members from diverse nationalities. Everyone was very cooperative, and we even worked together until midnight. My contribution was mainly organising the team members and following up on each member’s tasks. Additionally, I made a template of a poster using an online design service called Canva. Members, including myself, were not familiar with the tool, but we made an effort to get used to it and ended up with a good, consistent poster. From this assignment, the aspect that I was most interested in is the open-source software called OpenStack, which Catalyst Cloud is applying to its system. It’s worth exploring, as it may contribute to a deeper understanding of cloud infrastructure, since the major cloud companies do not reveal the technology behind the cloud service.

Collecting information about the NZ agency was challenging, especially since it was difficult to find how they are applying cloud technology to their business. However, we attempted to find some related information and discovered several reasons why the NZ agency is using Catalyst Cloud. This not only prioritises the local company but also has several compelling reasons for its choice. Finally, this research may have made us feel more familiar with New Zealand.

4. Hanjie Liu (1606580)

In this group project on Catalyst Cloud and the New Zealand government’s cloud

strategy, my main contributions focused on the chapters "Security and Governance Considerations" and "Cost and Scalability." I was responsible for researching, analyzing, and writing these chapters, ensuring they met New Zealand public sector standards such as NZISM, PSR, and ISO certifications, as well as Catalyst Cloud's documented controls. This required me to interpret a complex security framework and translate it into a clear, academic, and report-structured explanation.

In the "Cost and Scalability" chapter, I examined Catalyst Cloud's pricing model, financial operating practices, and resilience features. A key challenge was ensuring accuracy and credibility, especially given the lack of consistency or detail in some publicly available documentation. To address this, I cross-checked multiple sources, validated references, and carefully integrated them into the final writing. Refining technical details into concise and understandable content was another challenge, but it enhanced my ability to clearly communicate cloud concepts.

Working with a five-person team also highlighted the importance of coordination. We had to ensure consistency in formatting, writing style, and reference structure across all chapters. Regular discussions helped us unify our content, reduce duplication, and ultimately produce a clear, logically sound report.

In conclusion, this task deepened my understanding of cloud governance, cost optimization, and public sector cloud strategy. More importantly, it improved my research skills, technical writing abilities, and teamwork skills.

5. Nikeeta Kumari (1582117)

This group project was a good learning experience for me. I worked on several tasks that helped us finish the report, poster, and presentation on time. One of my main contributions was finding references. I searched for government documents, cloud policies, Catalyst Cloud information, and official guidelines. The biggest challenge I faced was finding peer-reviewed academic papers, because most information on this topic comes from government websites, not research journals. This taught me how to look for trusted sources even when academic papers are limited.

I also completed the Risks, Issues, and Recommendations section of our report. This involved understanding the main problems NZ Government faced in cloud adoption and suggesting simple and practical solutions. Later, I wrote the conclusion, summarising the full report in a clear way.

Another area where I contributed was helping with diagrams. I supported the team in creating visuals that worked well for both the report and the poster. A big challenge for me was compiling all sections in LaTeX. Every team member has their own writing style, structure, and diagram format. Bringing everything together into one clean document was difficult. I had to adjust formatting, fix layout issues, and make sure the writing looked consistent. Even though it took extra time, it helped me improve my editing and formatting skills.

For the presentation, I created the base PowerPoint layout. This gave everyone a

starting point so they could add their content easily. I also completed my own sections and made sure the title and agenda slides looked professional. Our team worked well together. We had regular meetings to talk about progress, share ideas, and solve problems. Everyone contributed equally, and we always helped each other when someone was stuck. From this project, I learned a lot about teamwork, communication, research, and design. I also improved my skills in LaTeX, diagram creation, and organising group work.